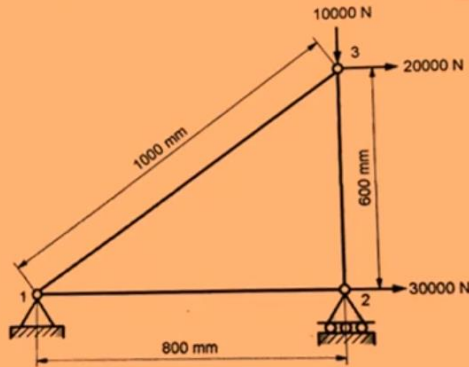


Practical 2

Title - Truss Analysis using 1D Element

Sample Example

Stress and deflection analysis of 2D truss (FEA)



$E = 200 \text{ GPa}$

$A = 300 \text{ mm}^2$

Find:

1. Nodal displacement
2. Stresses in each element
3. Reaction forces

SOLUTION:

$$U_1 = V_1 = 0 \text{ mm}; \quad U_2 = 0.4 \text{ mm}; \quad V_2 = 0 \text{ mm};$$

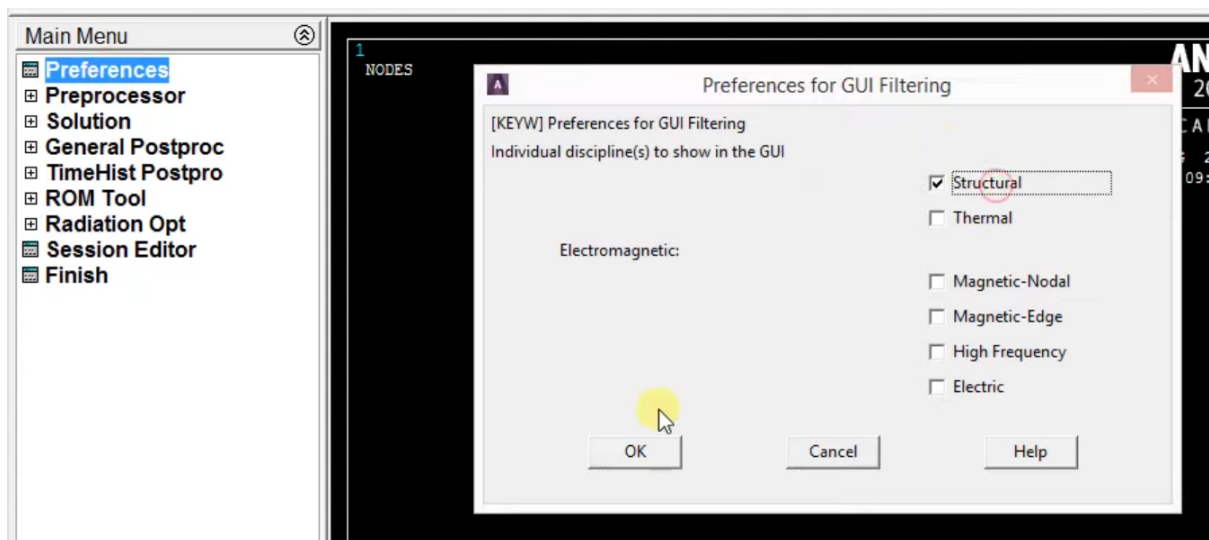
$$U_3 = 0.7083 \text{ mm}; \quad V_3 = -0.25 \text{ mm}$$

$$\sigma_1 = 100 \text{ N/mm}^2; \quad \sigma_2 = -83.33 \text{ N/mm}^2; \quad \sigma_3 = 83.33 \text{ N/mm}^2$$

$$R_{1x} = -50000 \text{ N}; \quad R_{1y} = -15000 \text{ N}; \quad R_{2y} = 25000 \text{ N}$$

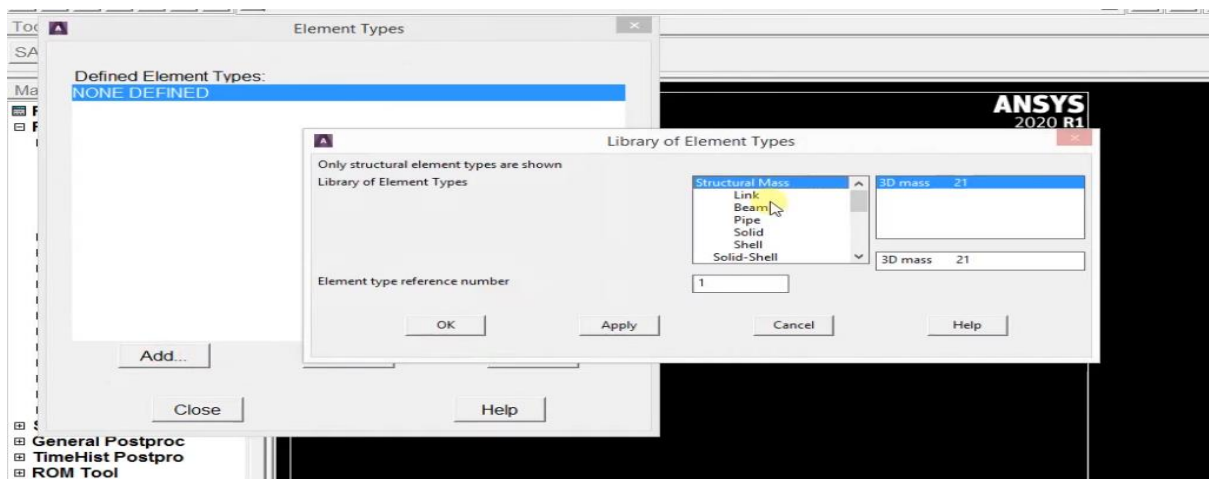
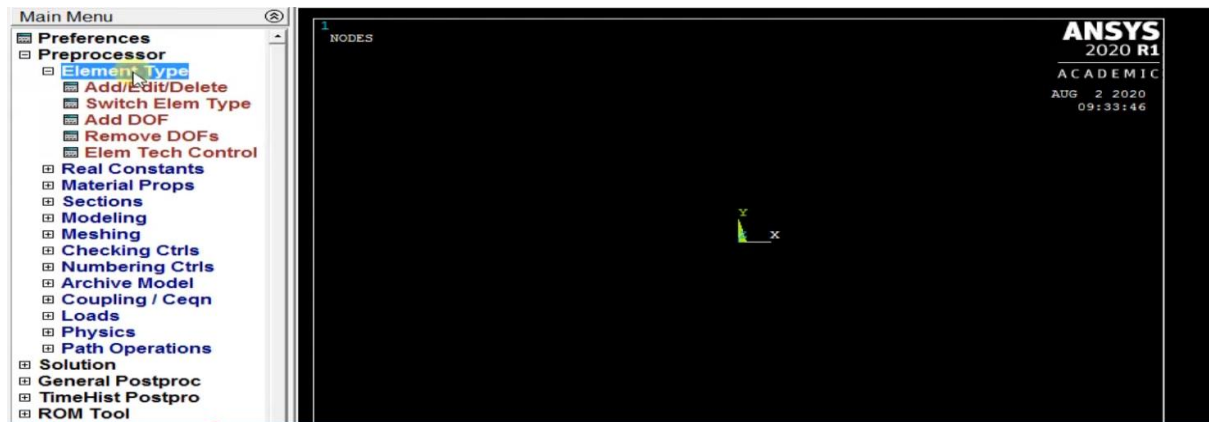
Step 1 Preferences

Open Ansys APDL – Preferences – Structural - OK



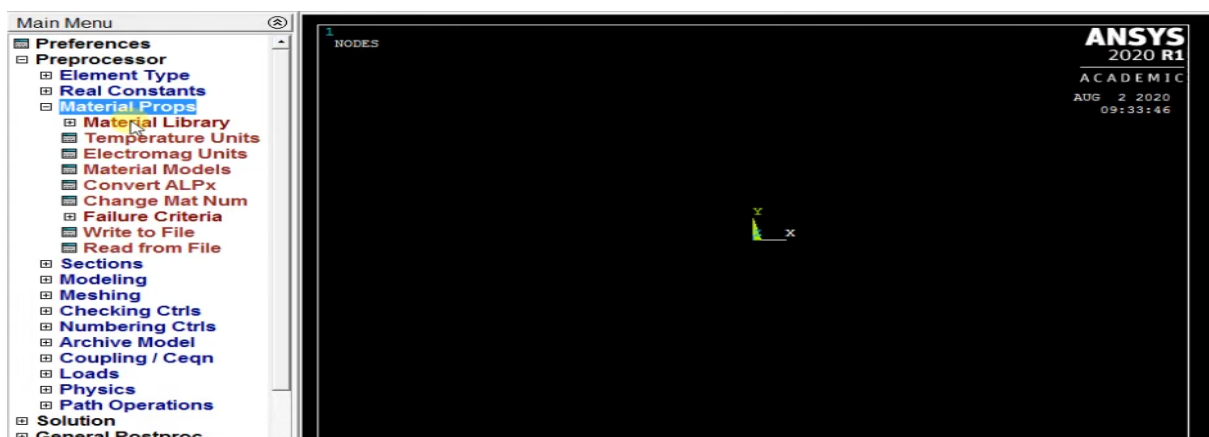
Step 2 Pre-processor

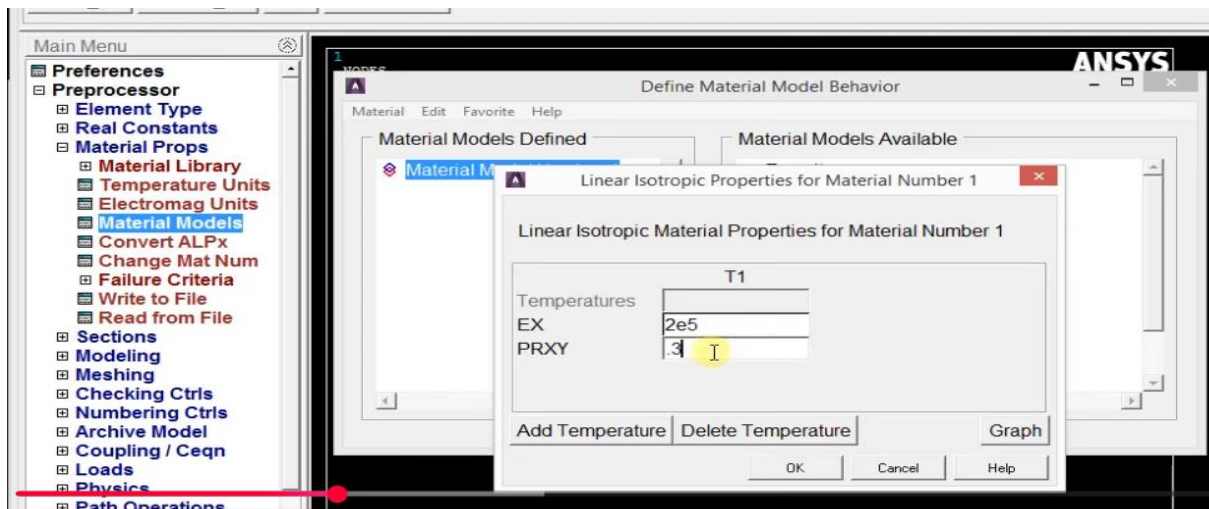
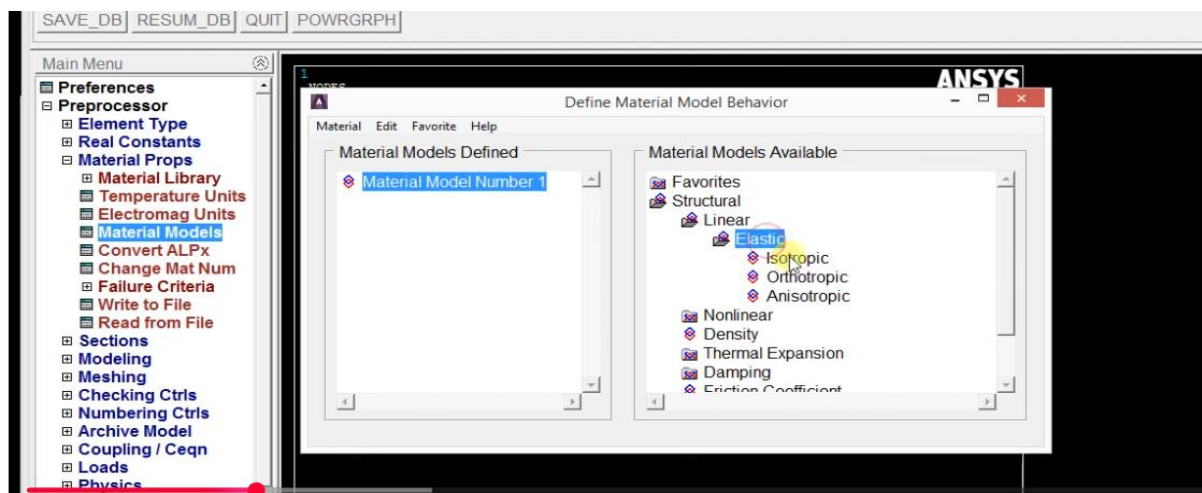
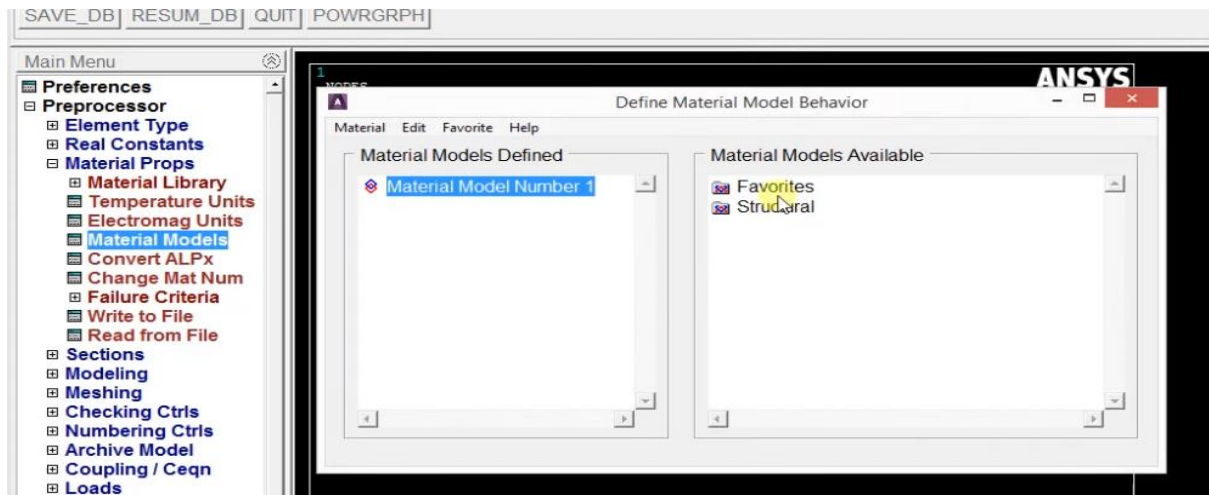
Pre-processor – Element Type – Add – Link –OK - Close



Step 3 Material

Material Properties – Material Models – Structural – Linear – Elastic – Isotropic – Enter Young's Modulus – Enter Poisson's Ratio - OK

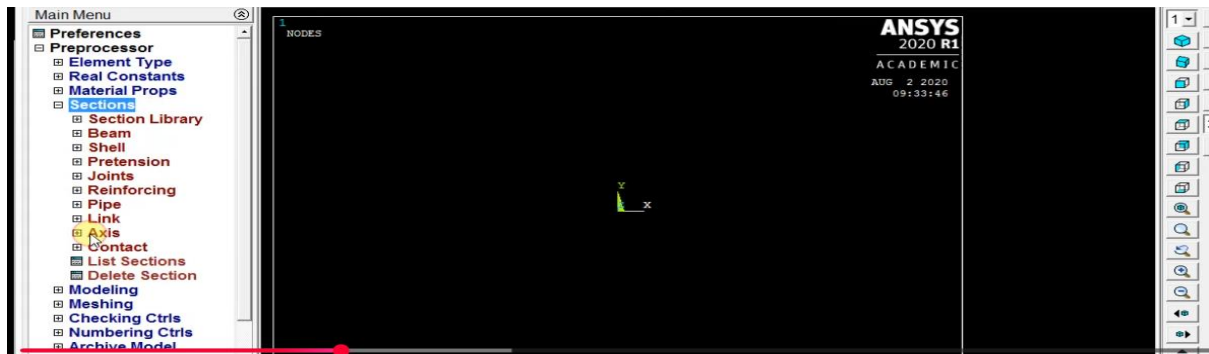




Step 4 Sections

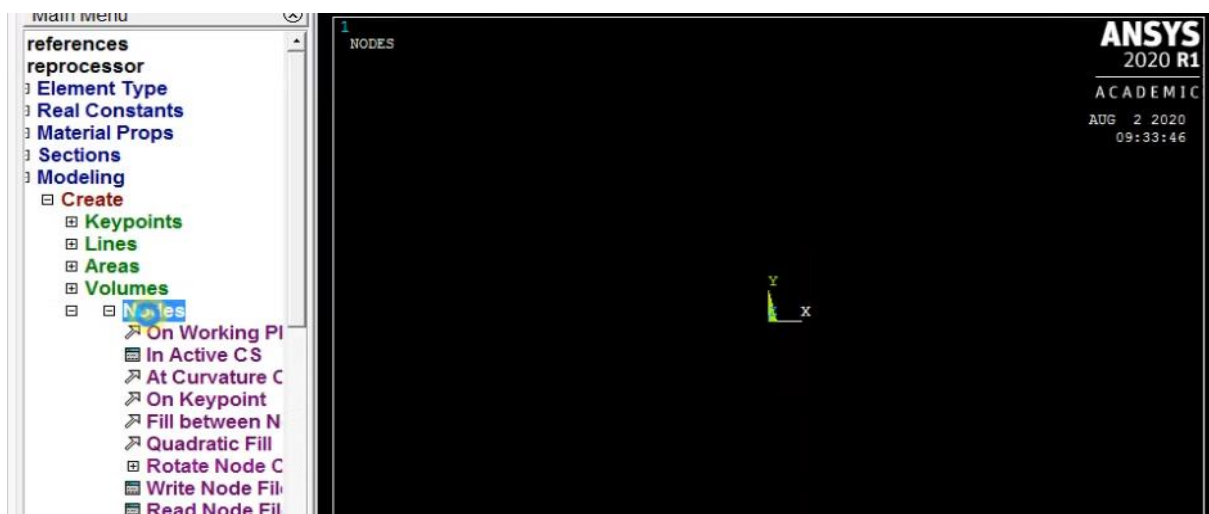
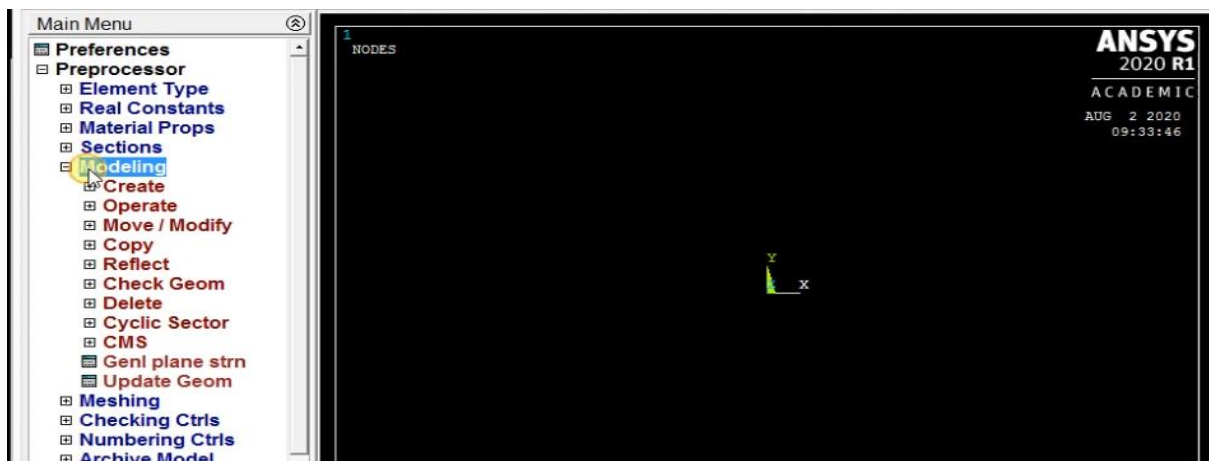
Sections – Link – Add – Link Section ID - 1 – OK – Enter Material Area –

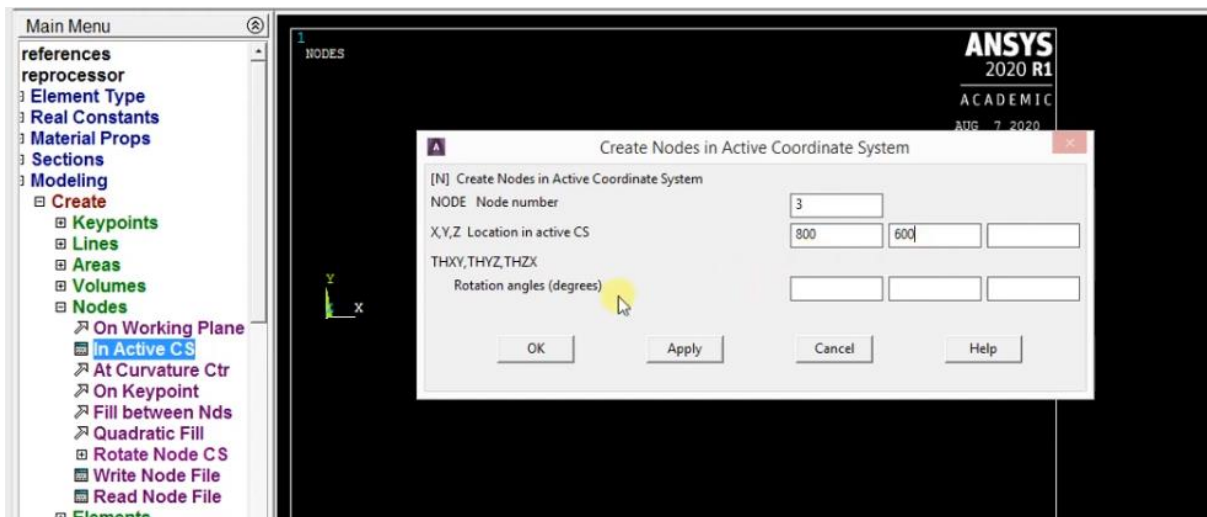
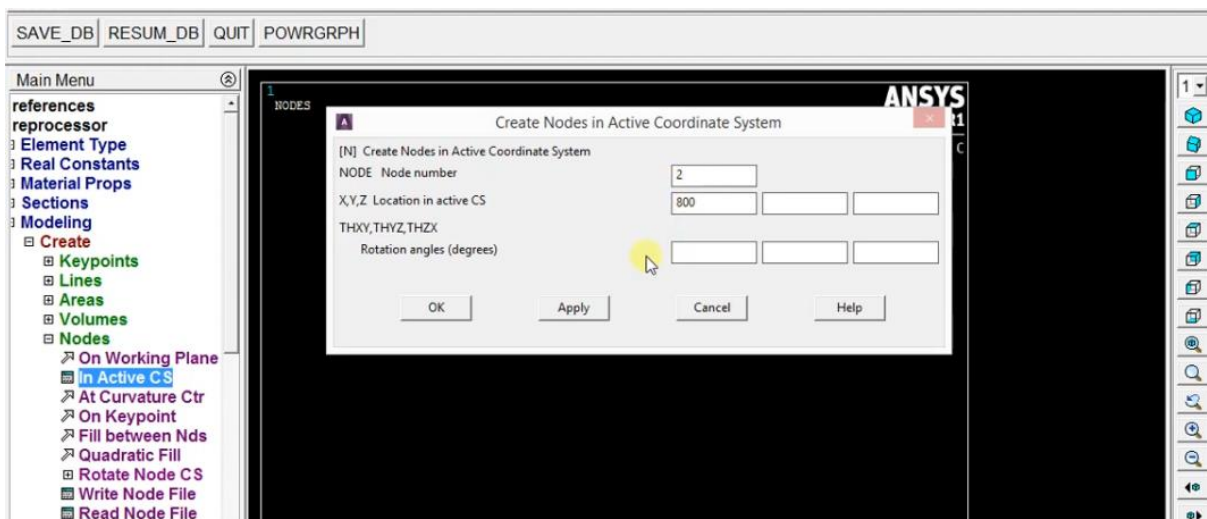
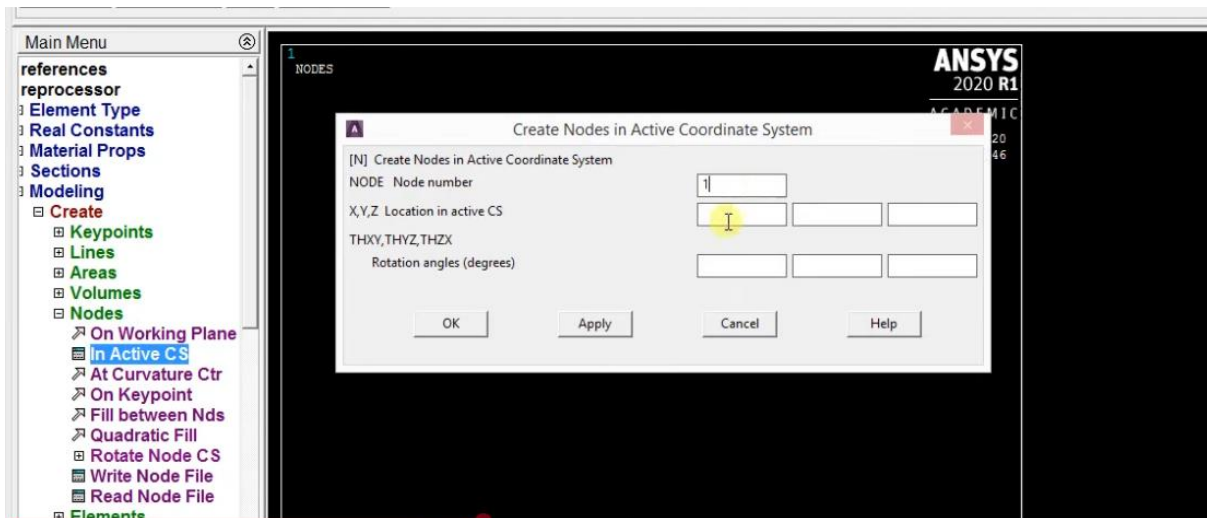
Apply - OK

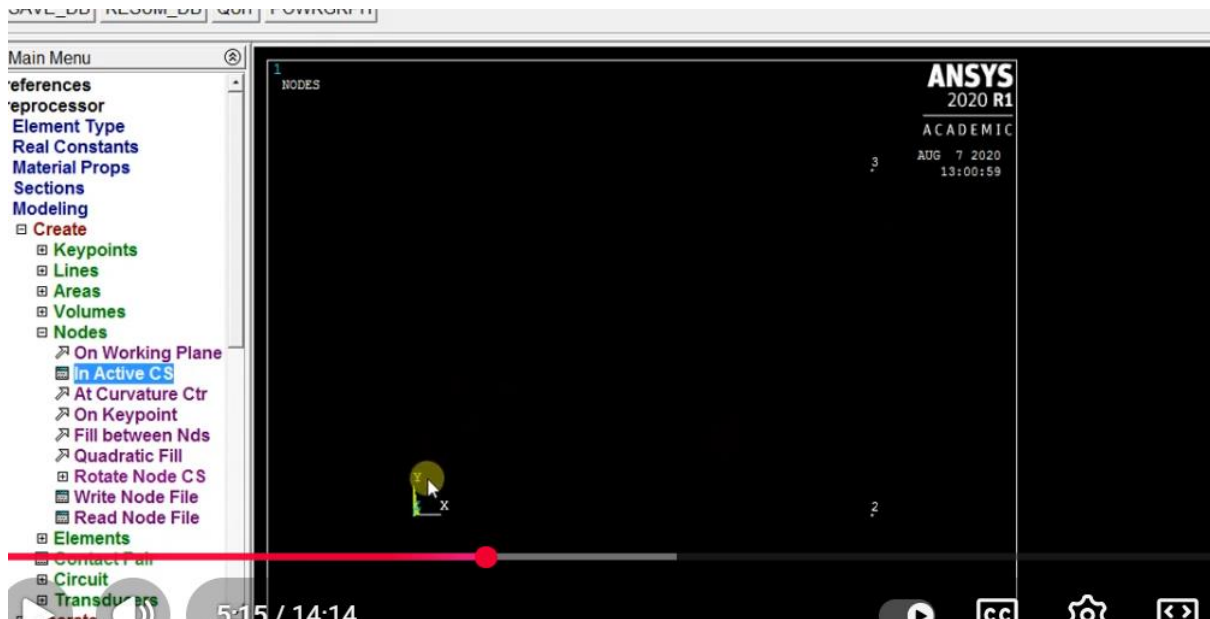


Step 5 Modeling

**Modeling – Create – Nodes – In Active CS – Node Number 1 – Location –
Apply - Node Number 2 – Location – Apply - Node Number 3 – Location –
OK**







Step 6 Joining of Nodes

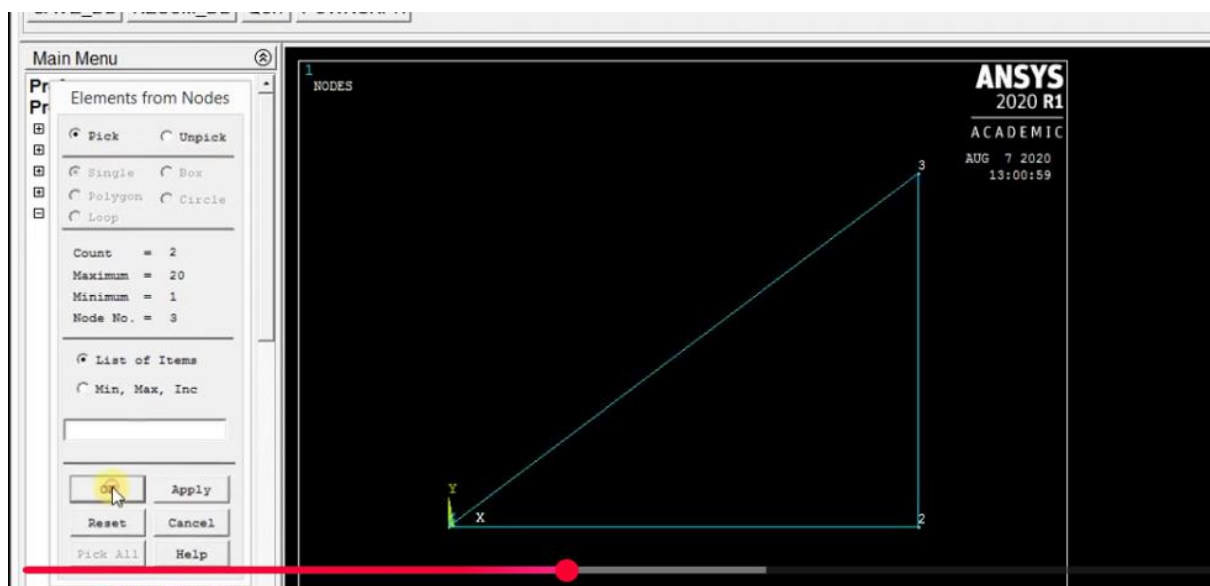
Modeling – Create – Elements – Thru Nodes – Select Node 1 to 2 – OK

For Second Element (Node 2 to 3)

Elements – Thru Nodes – Select Node 2 to 3 – OK

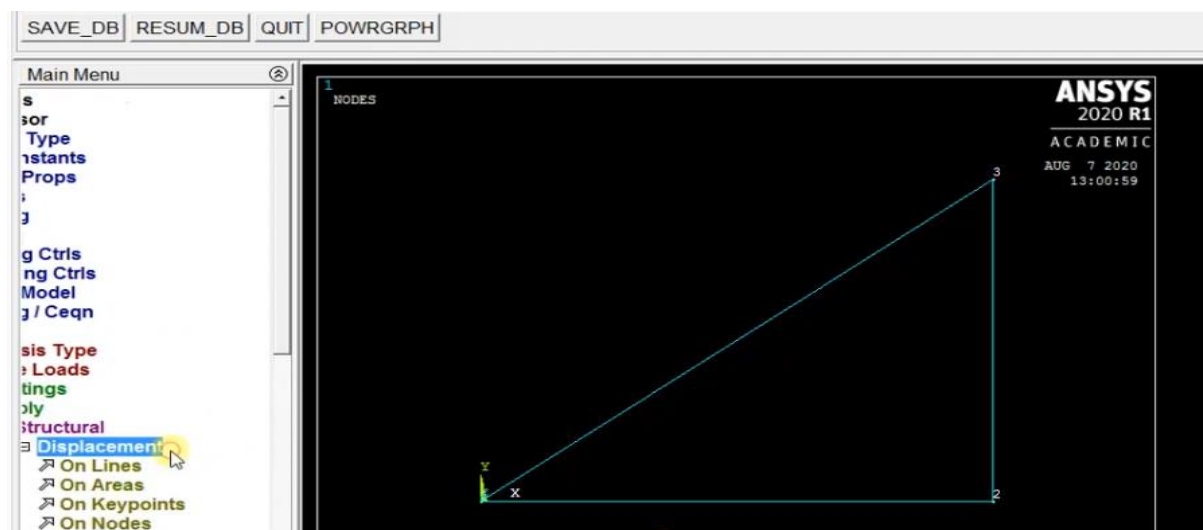
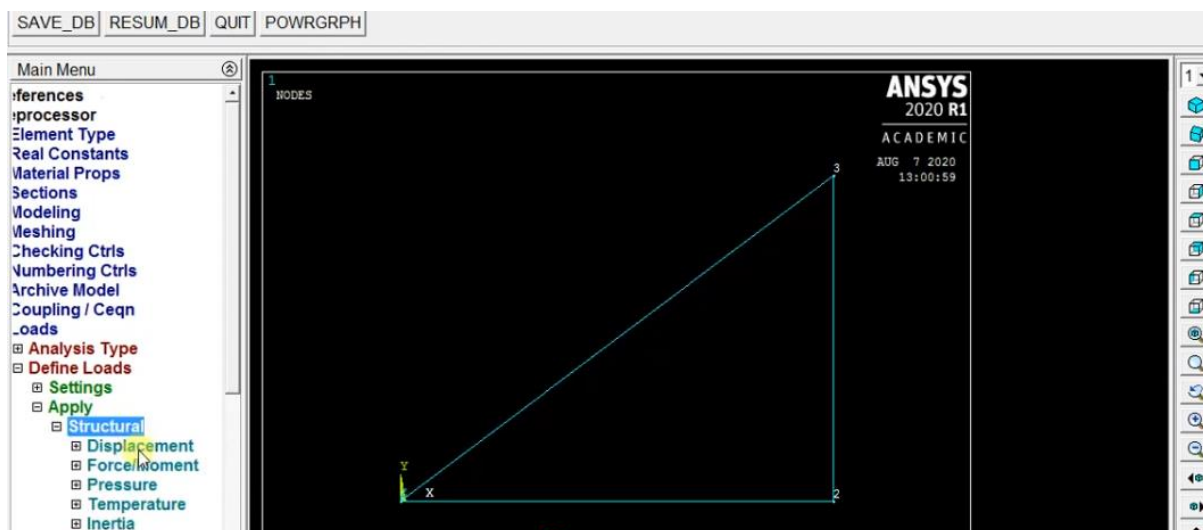
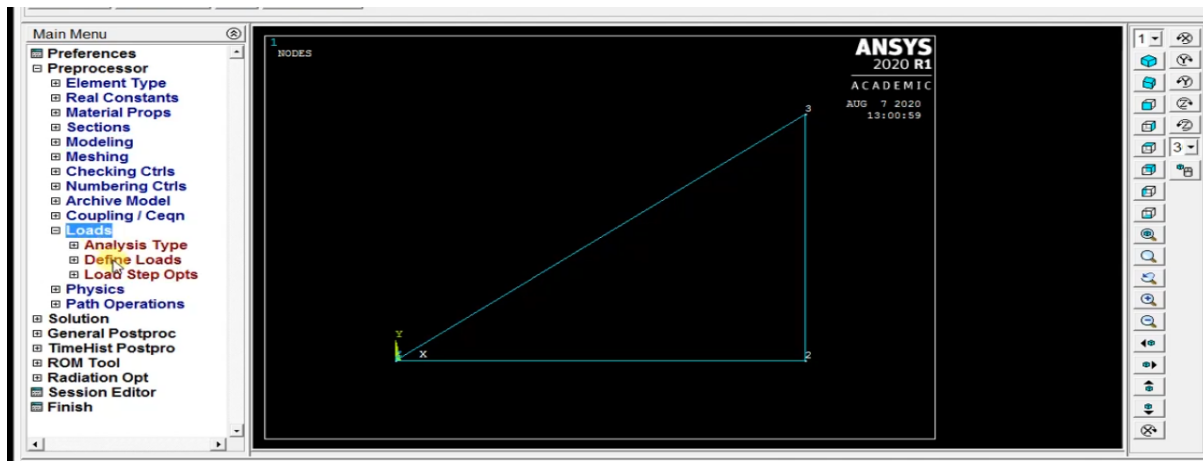
For Third (Node 1 to 3)

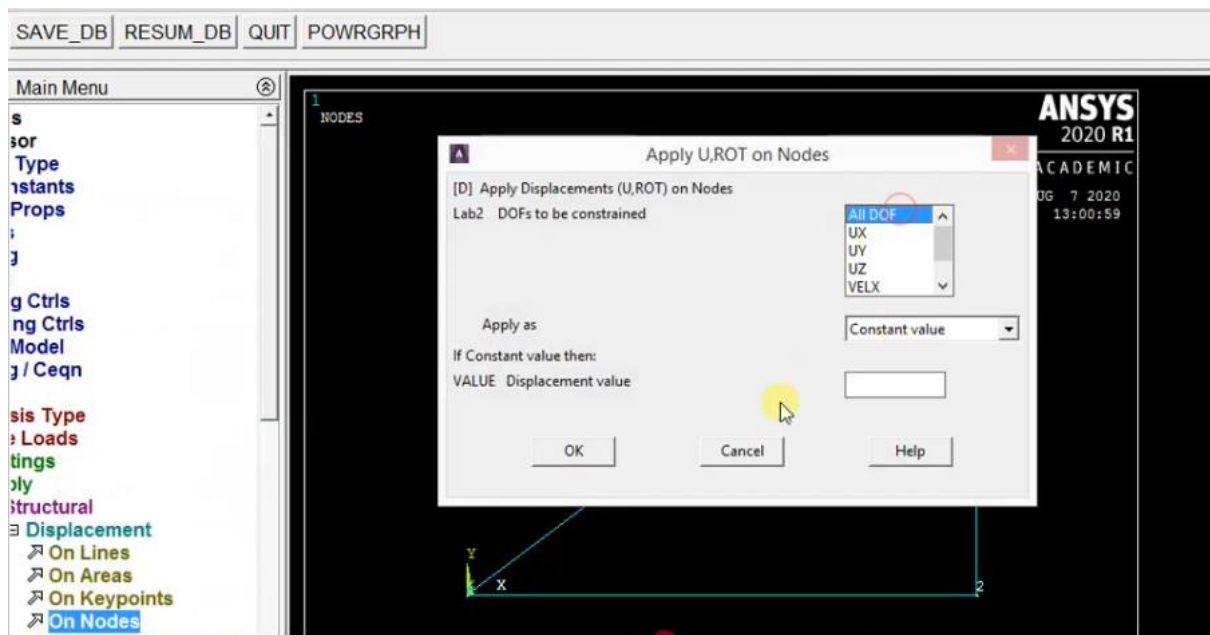
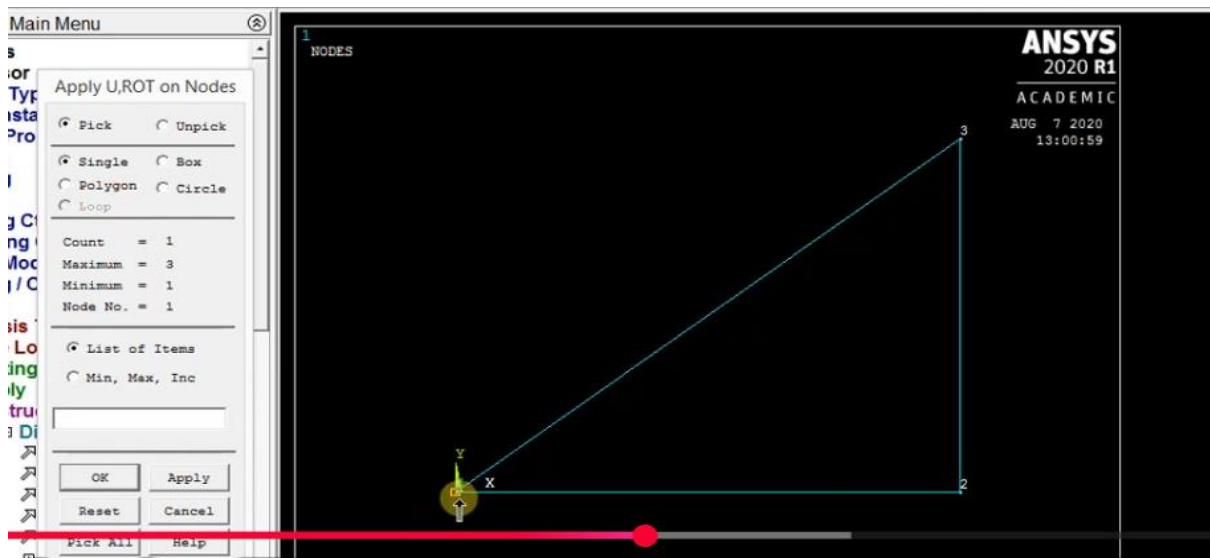
Elements – Thru Nodes – Select Node 2 to 3 – OK



Step 7 Apply Loads

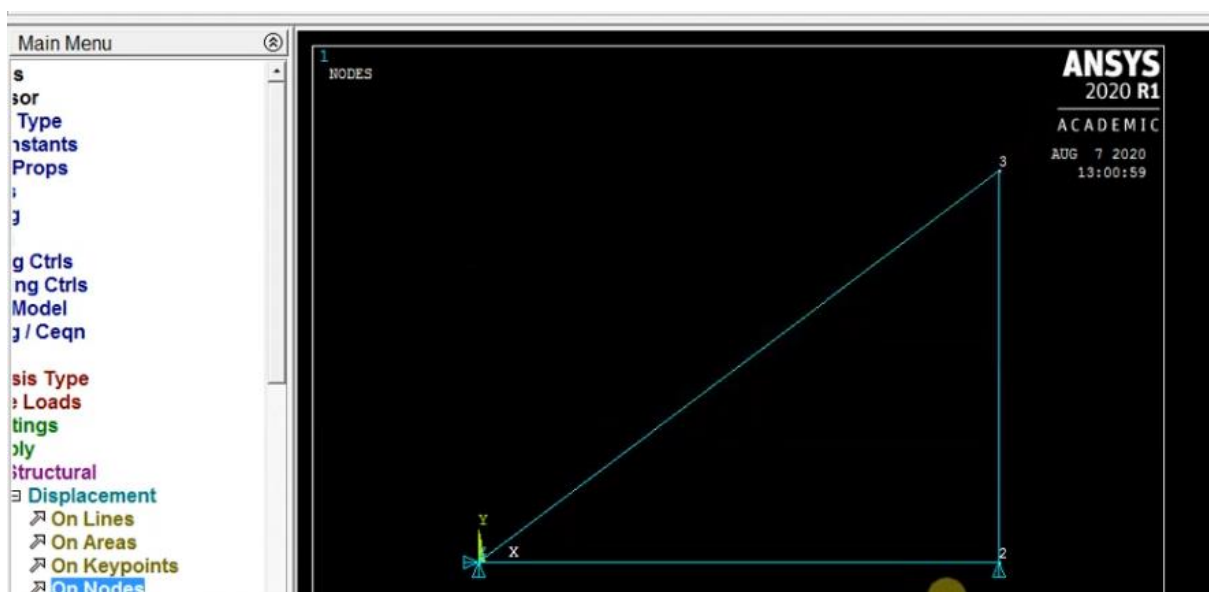
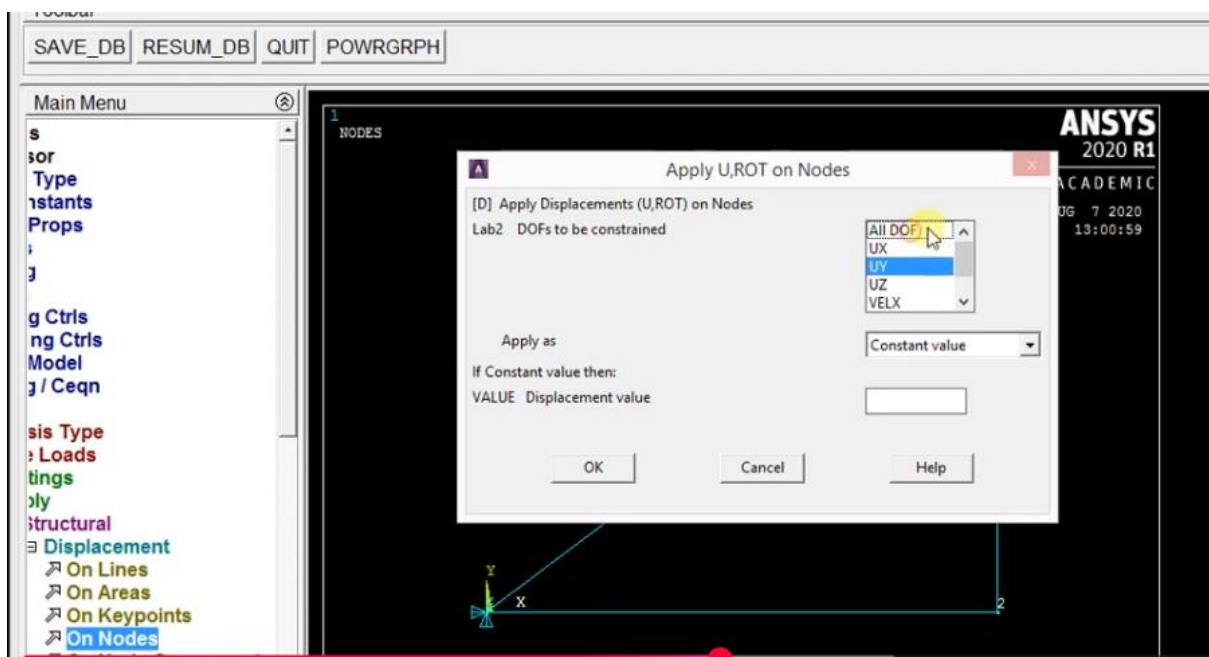
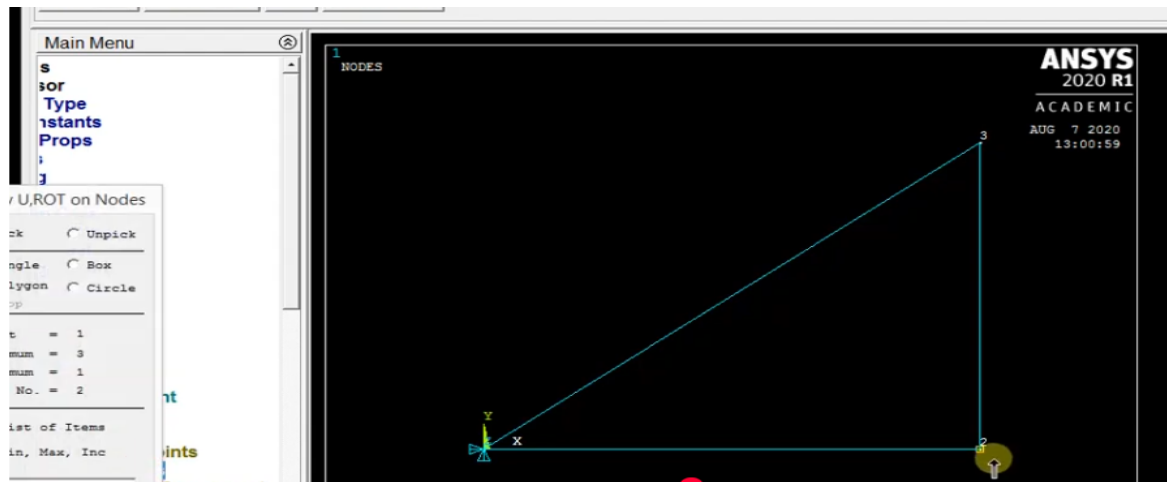
**Loads – Define Loads –Apply – Structural – Displacement – On Nodes –
Select 1st Node – All DOF – OK**





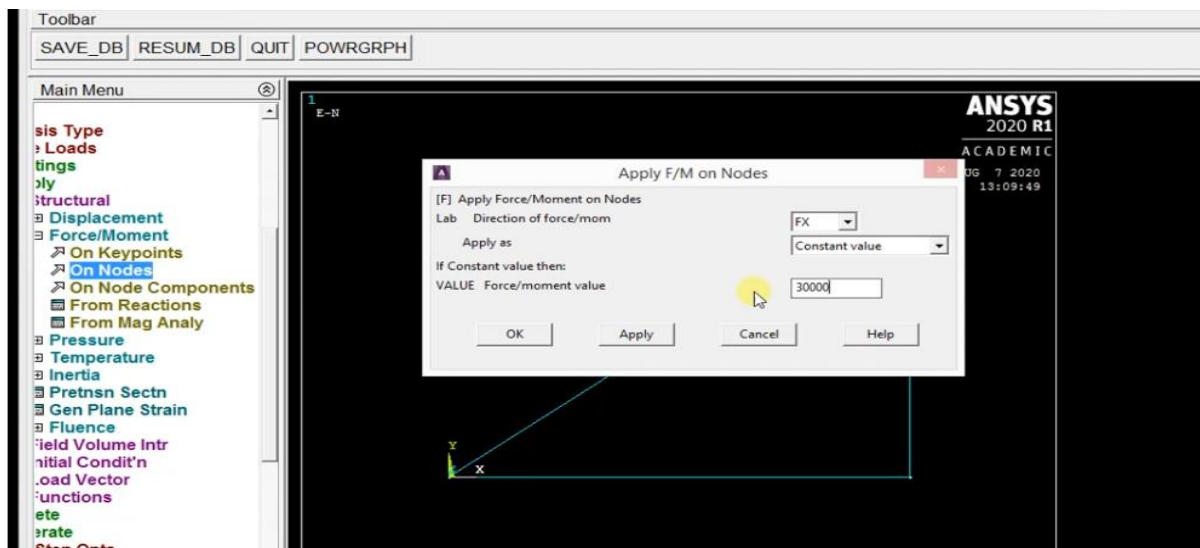
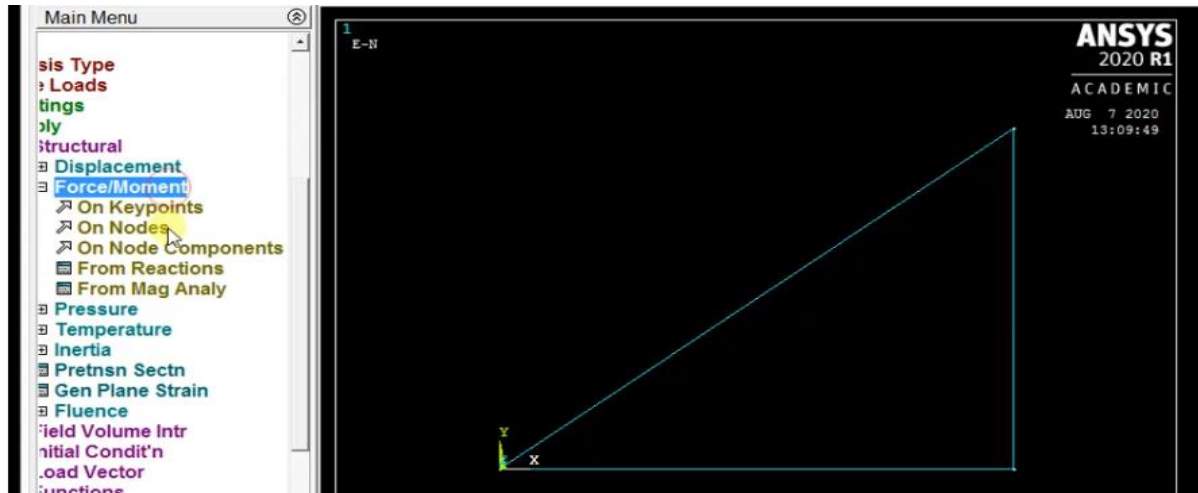
For 2nd Node

**Loads – Define Loads –Apply – Structural – Displacement – On Nodes –
Node – Select 2nd Node – UY – OK**



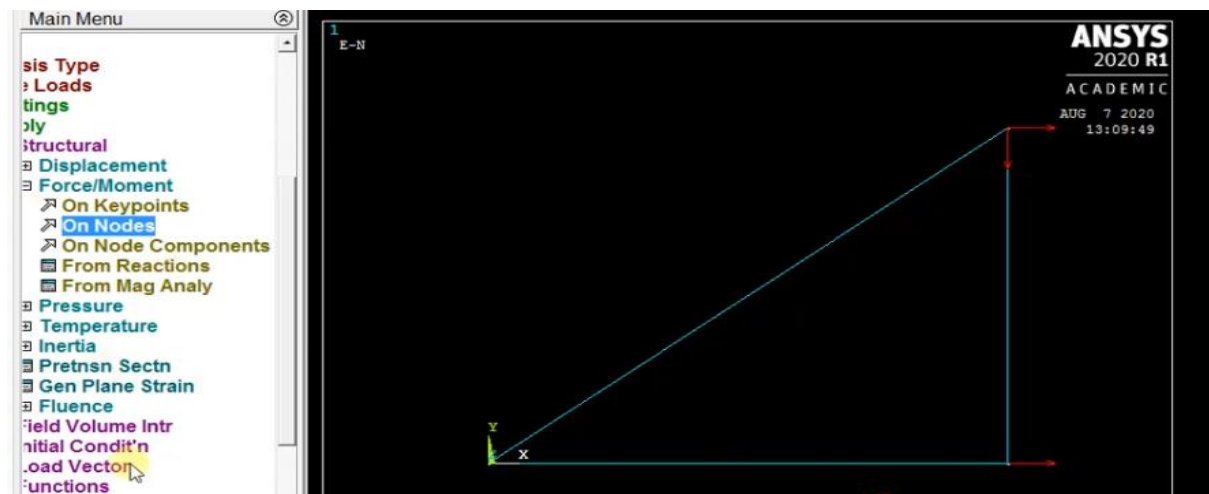
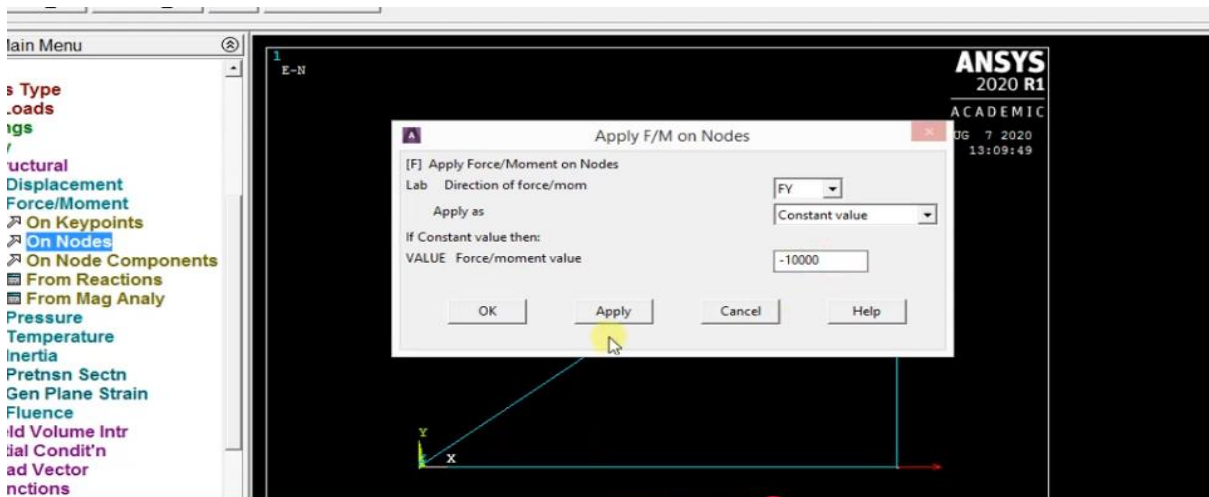
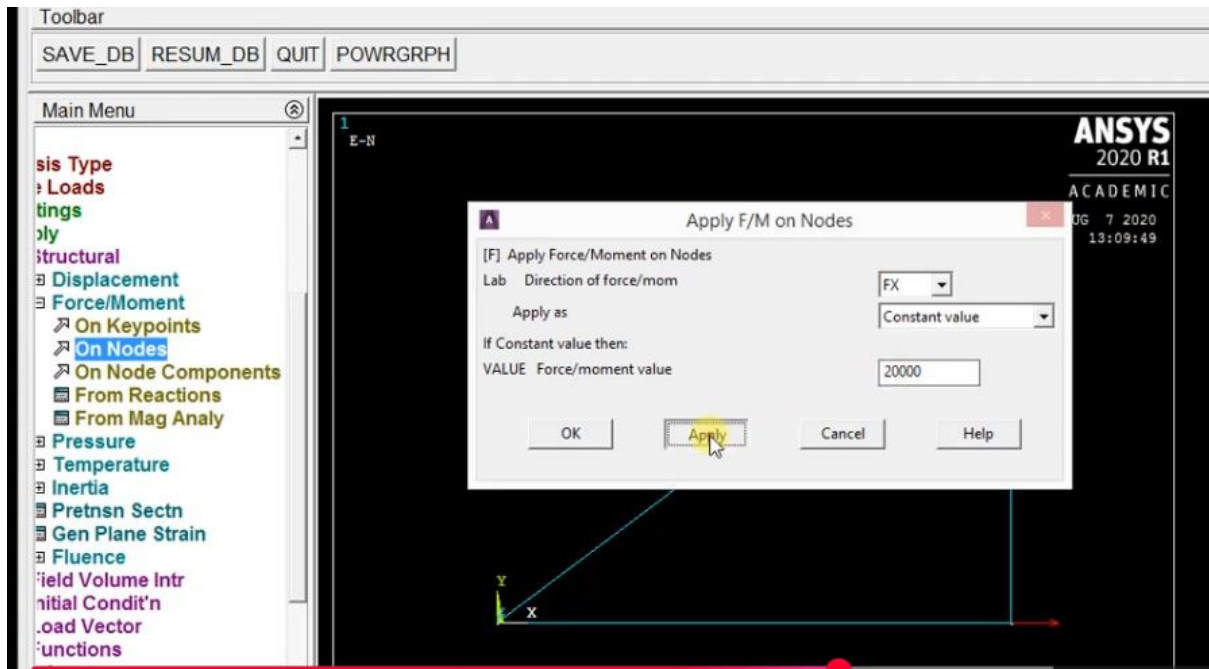
For 2nd Node

**Loads – Define Loads –Apply – Structural – Force – On Nodes –Node –
Select 2nd Node – FX – Enter Force Value - OK**



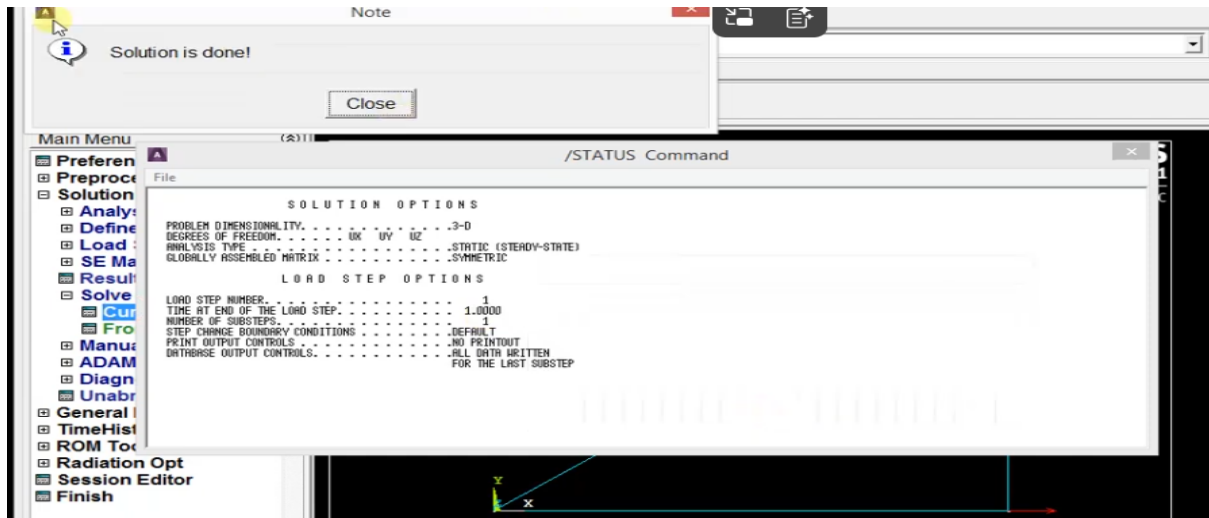
For 3rd Node

**Loads – Define Loads –Apply – Structural – Force – On Nodes –Node –
Select 3rd Node – FX – Enter Force Value - FY- Enter Force Value - OK**



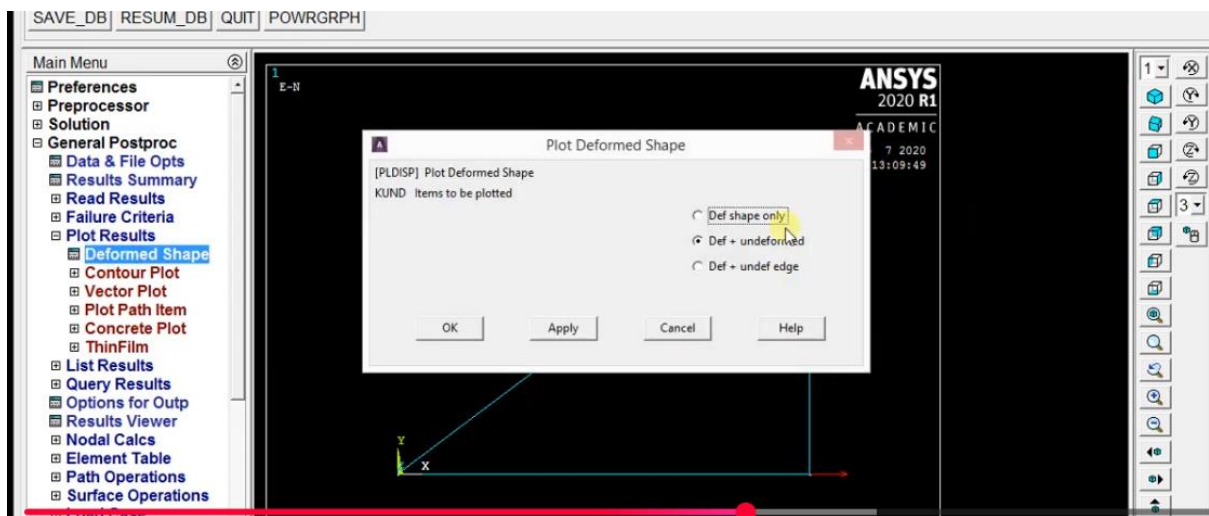
Step 8 Solution

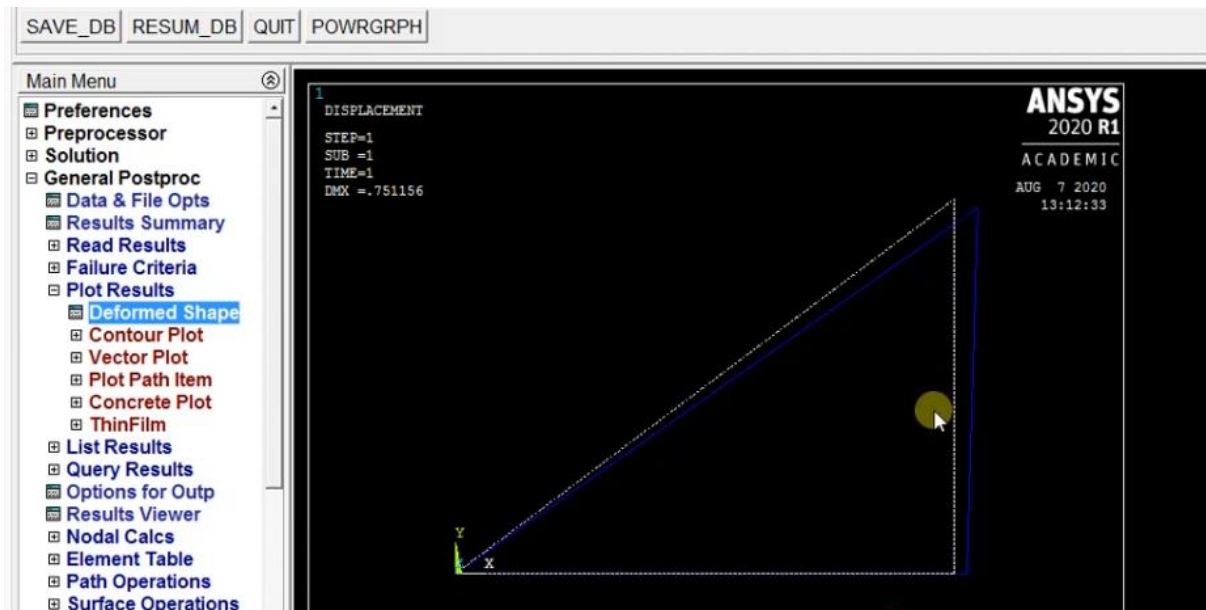
Solution – Solve – Current LS – OK



Step 9 General Results

General Post Process – Plot Results – Deformed + Undeformed – OK



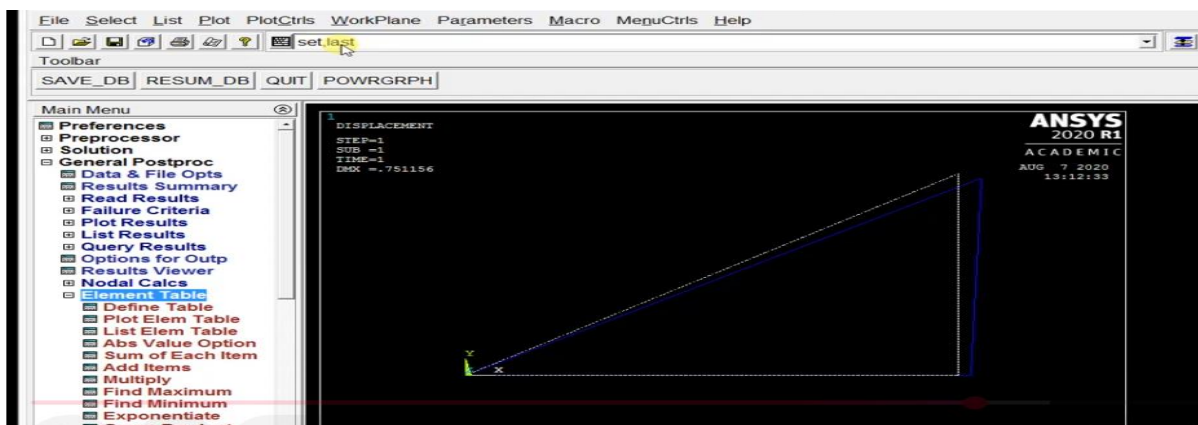


Step 10

General Post Process - Element Table

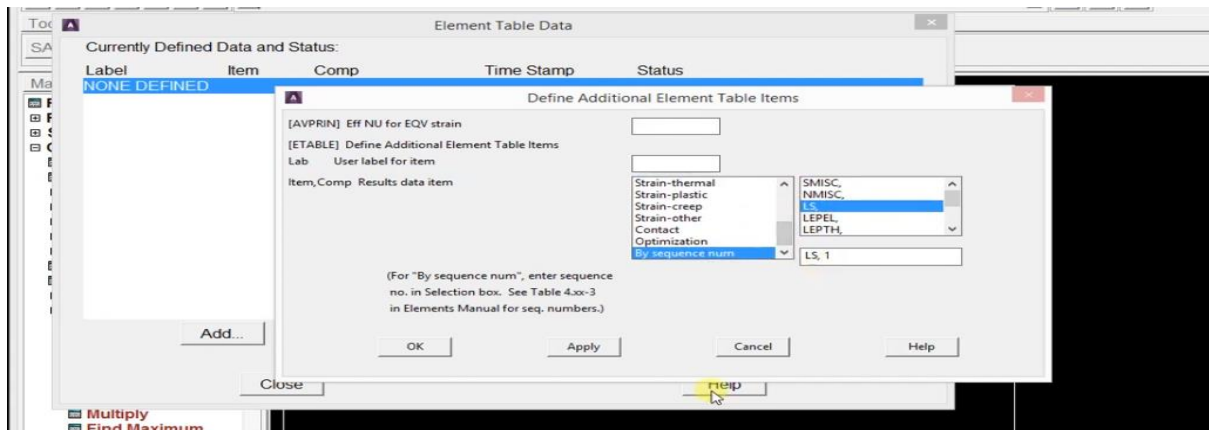
Step – 11 Command Prompt

Set, last



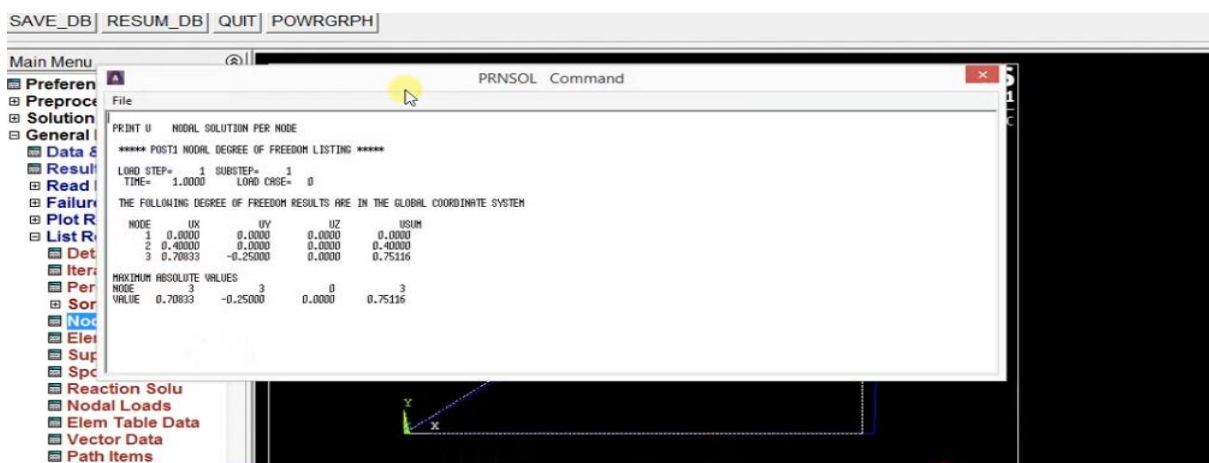
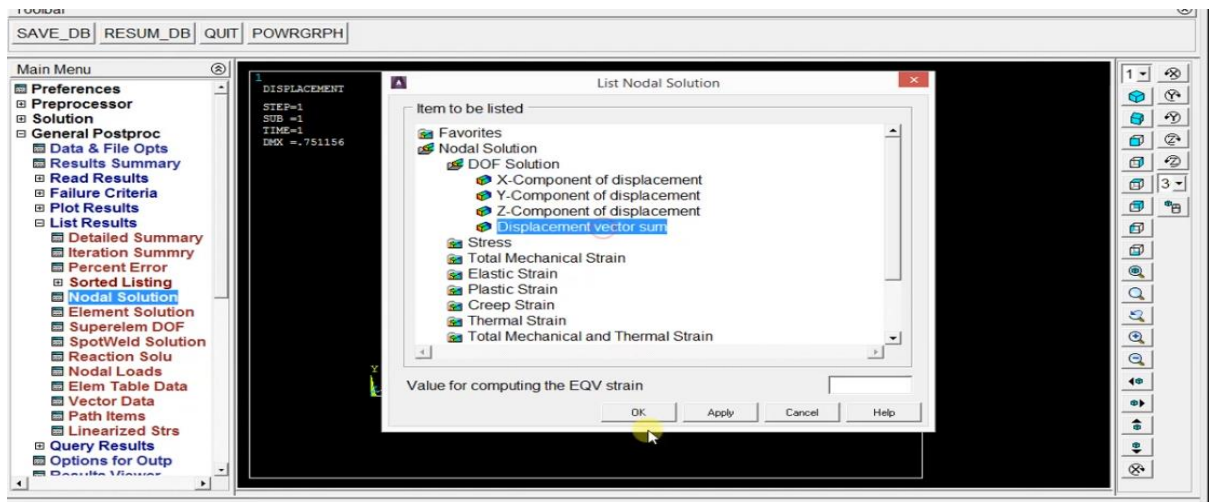
Element Table – Define Table – ADD – Select last (By sequence number) –

LS – LS, 1 – OK - Close

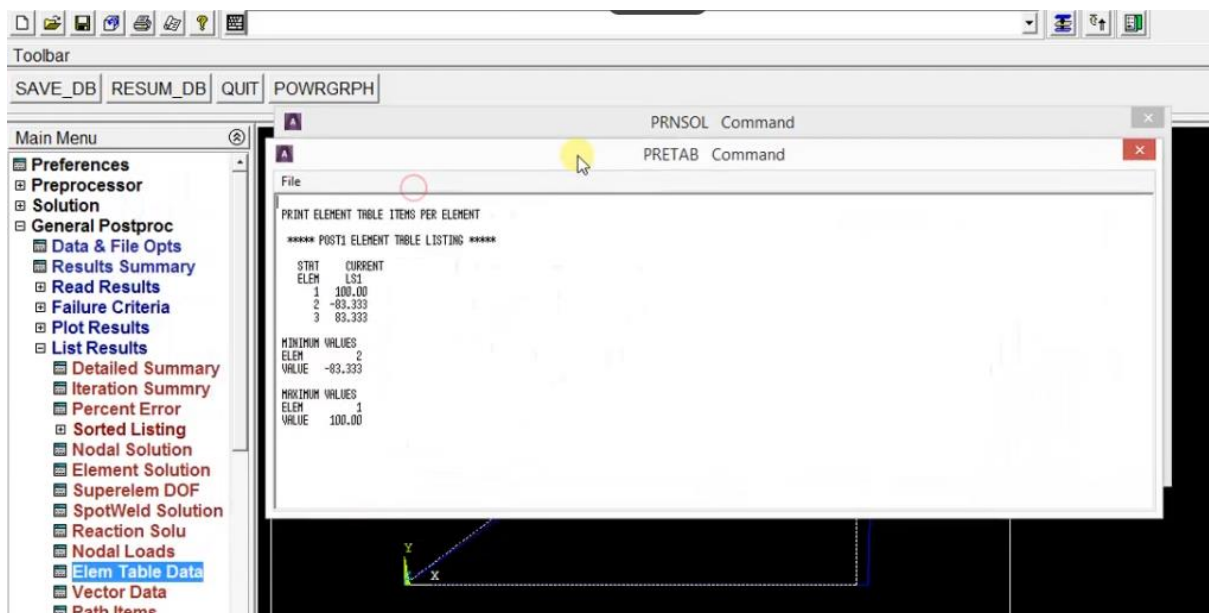
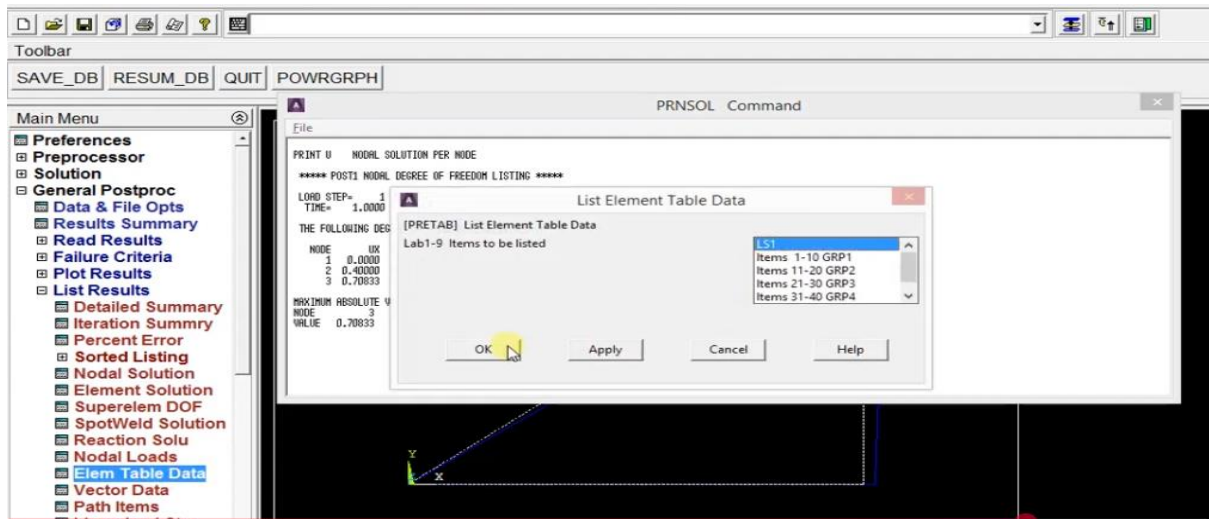


Step – 11 List Results

List Results – Nodal Solution – Displacement Vector Sum - OK

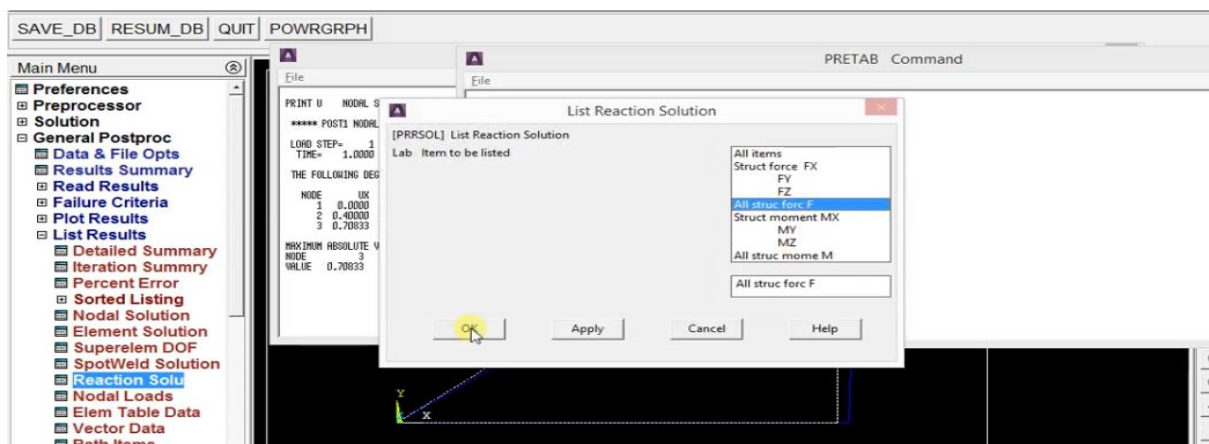


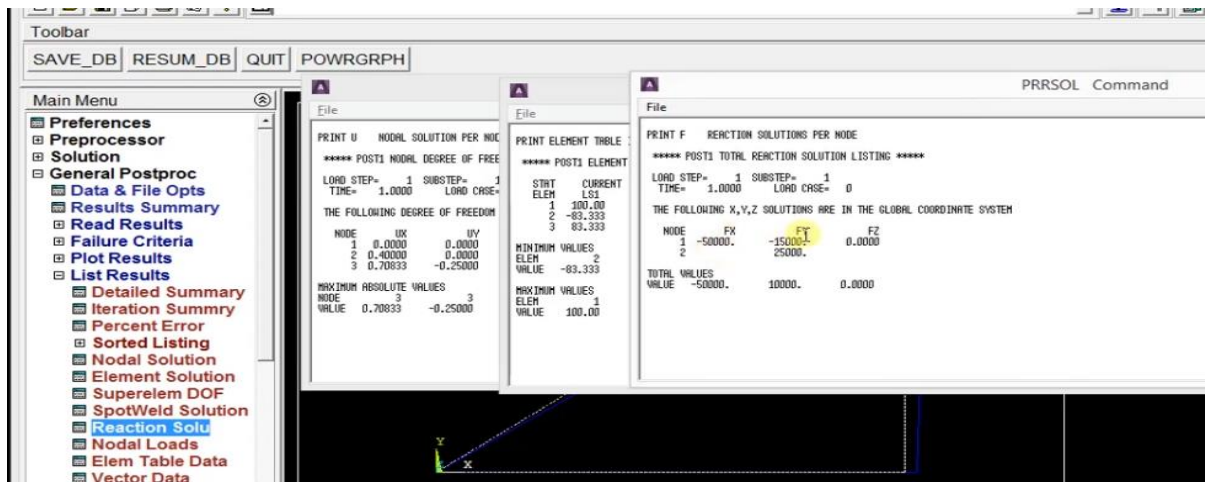
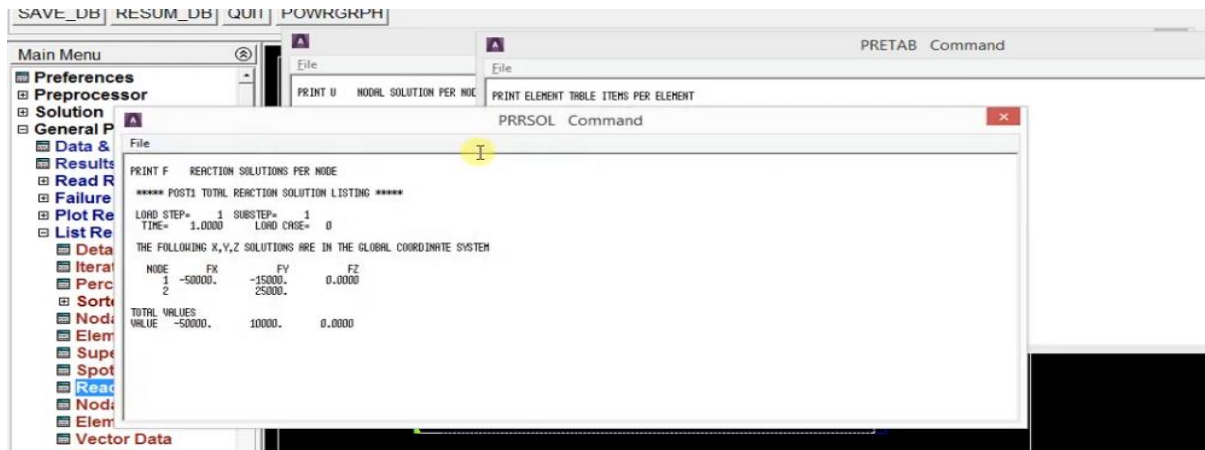
List Results – Element Table Data – LS1 - OK



To Check Reaction Solution

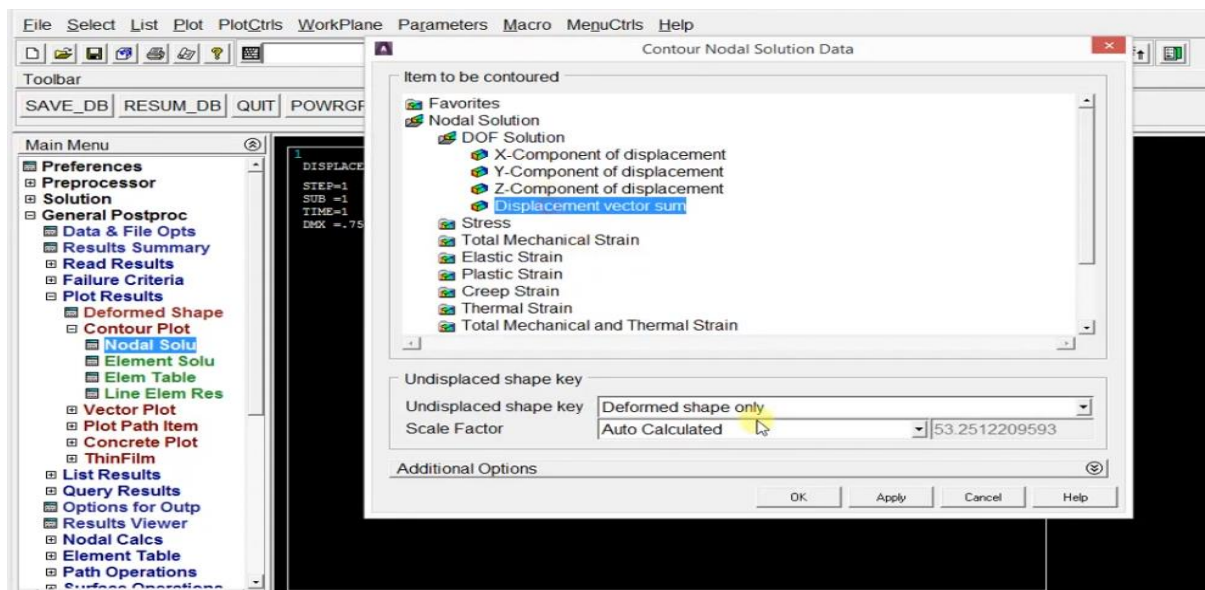
List Result - Reaction Solution – All Structural Force F - OK

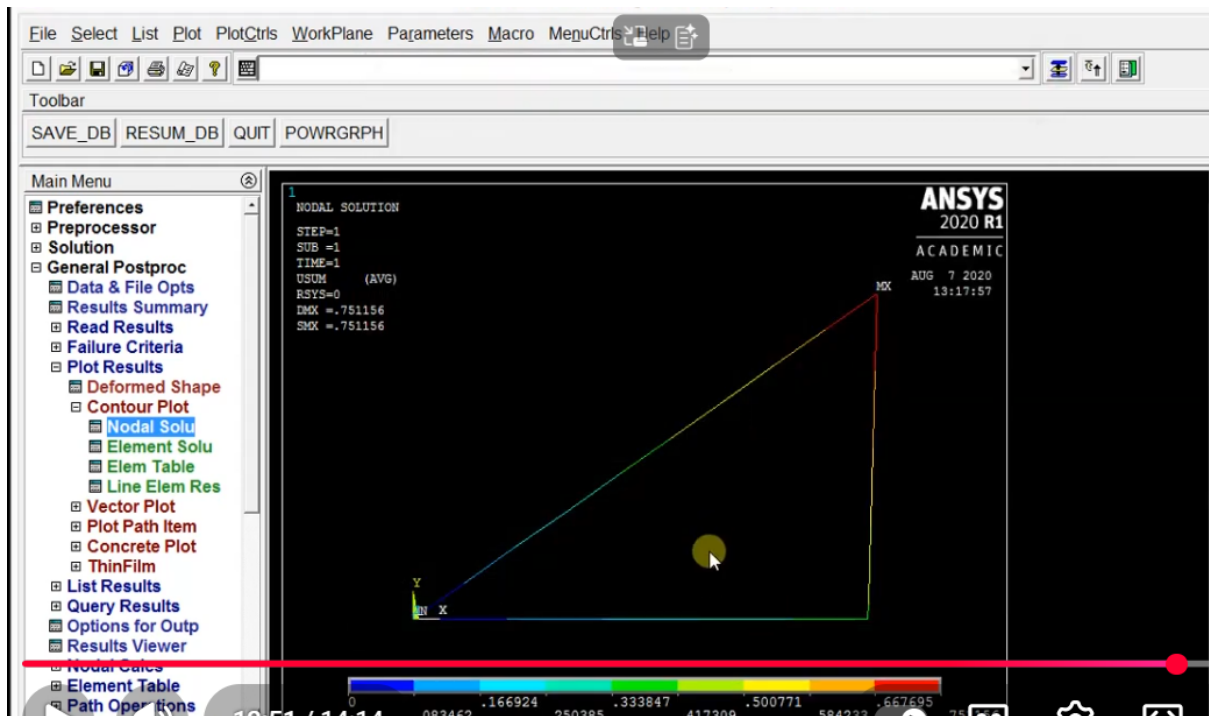




Step – 13 General Post Processing

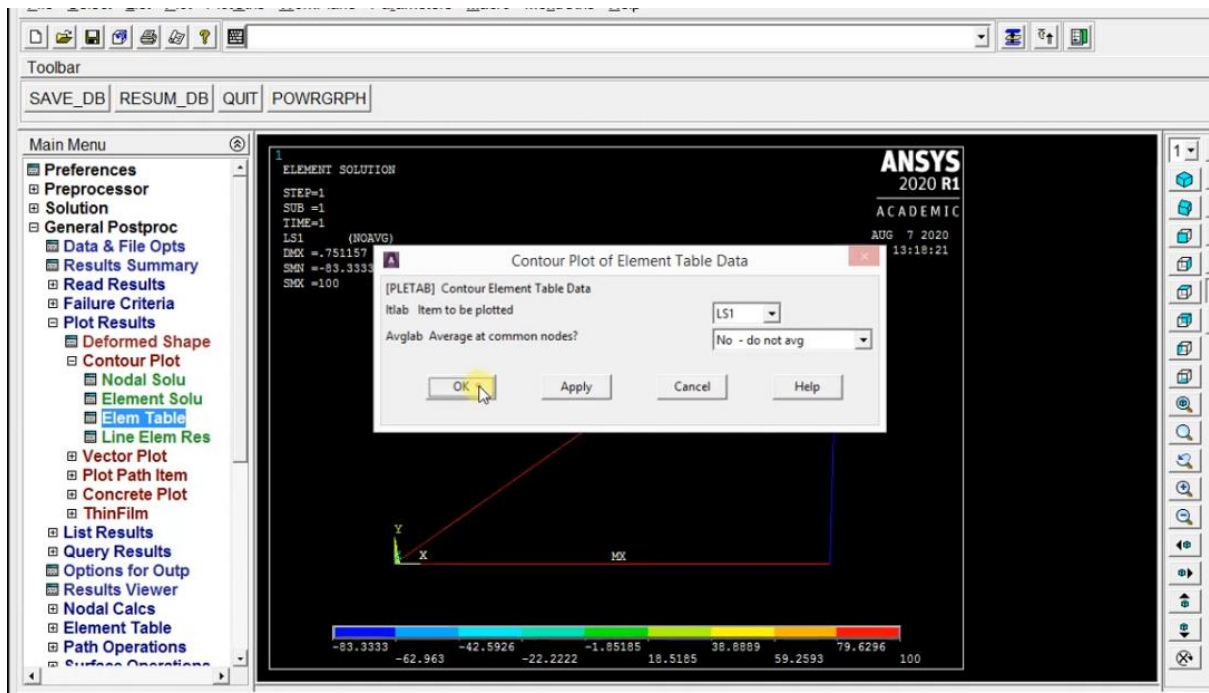
General Post Process – Plot Results – Contour Plot – Nodal Solution – Displacement Vector Sum – OK

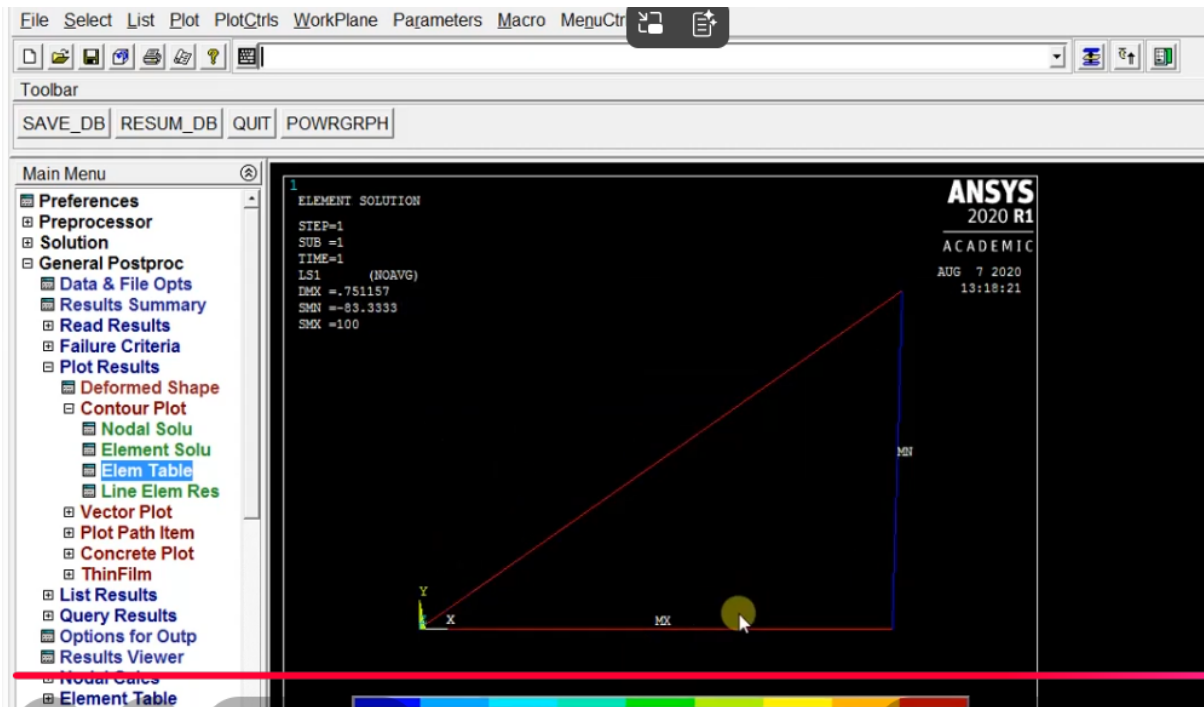




To See Stresses Plot

Plot Results – Element Table – LS1 - OK





To See Animation

Plot Controls – Animate – Deformed Shape – Deformed + un-deformed - OK

